🚀 10-Month Applied Data Science & AI Engineering Roadmap

Industry-Focused Learning Plan | From Data Foundations to LLM Systems

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👤 Prepared By

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📅 Program Duration

10 Months (Structured Learning & Build Track)

🎯 Program Objective

To build an **industry-ready Data Scientist / AI Engineer profile** through practical, project-driven learning focused on real-world problem solving instead of theory-only study.

This roadmap emphasizes:

✔ Hands-on implementation
✔ End-to-end AI systems
✔ Production mindset
✔ Open-source collaboration
✔ Portfolio-first learning

🧭 Learning Architecture (High-Level)

Phase 1 — Data Foundations

Month 1–2
Python • Pandas • SQL • Data Cleaning • Business Analysis

Phase 2 — Machine Learning Core

Month 3–4
ML Algorithms • Model Evaluation • Hyperparameter Tuning • Optimization

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Phase 3 — Applied Data Science Engineering

Month 5–6
Feature Engineering • Pipelines • Model Deployment • API Integration

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Phase 4 — Deep Learning & Vision AI

Month 7–8
Neural Networks • PyTorch • CNN • Transfer Learning

Phase 5 — NLP & Modern AI Systems

Month 9–10
Transformers • LLMs • RAG Systems • Open Source Contribution

🛠️ Core Skill Stack

- Python & SQL
- Machine Learning (Scikit-learn)
- Deep Learning (PyTorch)
- NLP & Transformers
- API Development (FastAPI)
- LLM Integration & RAG
- GitHub & Open Source Workflow

🏗️ End Deliverables

✔ Production-style AI projects
✔ Deployable ML APIs
✔ LLM-based AI Assistant (Capstone)
✔ Open Source Pull Requests
✔ Industry-ready GitHub Portfolio

🎯 Final Career Outcome

Target Roles:

- Data Scientist
- Machine Learning Engineer
- Applied AI Engineer
- LLM / GenAI Developer

⚡ Guiding Principle

Learning by Building — Not by Watching

🎯 Month-1 Goal (clear target)

Month end tak tum:

✔ Python basics comfortable
✔ Data clean kar pao
✔ CSV/Excel data read & analyze kar pao
✔ Basic graphs bana pao
✔ 1 mini analysis project GitHub pe upload karo

📅 Week-by-Week Plan

🗓️ Week 1 — Python Foundations (DS version)

Focus: **general coding nahi — data science wala Python**

Topics

- Variables, data types
- Lists, dictionaries, tuples
- Loops & conditions
- Functions
- Basic file handling

Practice

Daily small problems:

- numbers filter karna
- list sort karna
- dictionary count

✅ Checklist (Week 1)

- Python install + VS Code / Jupyter setup
- 20 small Python exercises
- Functions likhna samajh aa gaya
- CSV file read kar li (basic)

🗓️ Week 2 — NumPy + Pandas Start (MOST IMPORTANT)

Ye data scientist ka asli weapon hai.

NumPy

- Arrays
- Indexing
- Basic operations

Pandas

- DataFrame
- read_csv()
- head(), info(), describe()
- column select karna

Practice:
Random Kaggle dataset le lo aur explore karo.

✅ Checklist (Week 2)

- Dataset load kar pa rahe ho
- Columns filter kar pa rahe ho
- Missing values identify kar pa rahe ho
- Basic stats nikal pa rahe ho

🗓️ Week 3 — Data Cleaning (REAL SKILL)

Industry me 70% kaam yahi hota hai.

Topics

- Missing values handle karna
- Duplicate remove karna
- Data types change karna
- Basic transformations

Practice:
Dirty dataset lo and clean karo.

✅ Checklist (Week 3)

- Null values handle ki
- Duplicate rows remove ki
- Data type fix ki
- Clean dataset save kiya

📅 Week 4 — Visualization + Mini Project

Ab data bolna chahiye 📊

Visualization

- Line chart
- Bar chart
- Histogram
- Scatter plot

Tools:

- Matplotlib
- Seaborn (basic only)

Mini Project (IMPORTANT 🔥)

Example:

- Netflix data analysis
- Sales analysis
- IPL dataset analysis
- Startup funding analysis

What to include:

1. Dataset intro
2. Cleaning steps
3. Key insights
4. Graphs
5. Conclusion

✅ Checklist (Week 4)

- 4–5 graphs banaye
- Insights likhe
- GitHub repo create kiya
- README likha (simple explanation)

📦 Month-1 Final Deliverable (VERY IMPORTANT)

GitHub pe ek repo ho:

```
Data-Analysis-Month1
│
├── data/
├── notebook.ipynb
├── cleaned_data.csv
└── README.md
```

🧠 Daily Routine (best schedule)

अगर daily 2–3 hrs:

- 45 min learning
- 60 min practice
- 30 min revision / notes

Consistency > speed.

⚠️ Common Mistakes (avoid karna)

❌ 10 courses dekhna
❌ Sirf videos dekhna
❌ Perfect banane me time waste

✔ 1 dataset → complete analysis

🔥 BONUS (secret tip)

Month-1 end pe tum LinkedIn ya GitHub pe post likho:

> “Completed my first real data analysis project”

Trust me — motivation double ho jayega 🙂

🎯 **“Data manipulation + SQL thinking + deeper analysis”**
Yahi phase tumhe beginner se **serious data person** banata hai.

🎯 Month-2 Goal (clear target)

Month end tak tum:

✔ SQL confidently likh pao
✔ Advanced Pandas use kar pao
✔ Multiple datasets join kar pao
✔ Real business-style analysis kar pao
✔ 1 strong analysis project GitHub pe upload karo

📅 Week-by-Week Plan

🗓️ Week 1 — SQL Basics (Data Scientist ke liye)

Data science interviews me SQL almost mandatory hota hai.

Topics

- SELECT
- WHERE
- ORDER BY
- LIMIT
- COUNT / SUM / AVG

Practice:
Small tables banake queries run karo.

✅ Checklist (Week 1)

- SQL environment setup (SQLite / PostgreSQL)
- Basic queries likh li
- Filtering samajh aa gaya
- Aggregation use ki

🗓️ Week 2 — SQL Intermediate (REAL INTERVIEW LEVEL)

Ab fun start 😄

Topics

- JOINS (INNER, LEFT)
- GROUP BY
- HAVING
- Subqueries

Practice example:

- Customers + Orders tables join karna.

✅ Checklist (Week 2)

- 20+ join queries practice
- Group by se insights nikale
- Subquery likh pa rahe ho
- Business question solve kiya

🗓️ Week 3 — Advanced Pandas (GAME CHANGER)

Ab Python + SQL mindset combine hoga.

Topics

- merge() (joins like SQL)
- groupby()
- pivot tables
- apply() functions
- sorting & filtering advanced

Practice:
2 alag CSV datasets merge karo.

✅ Checklist (Week 3)

- Dataset merge ki
- Groupby analysis ki
- Pivot table banayi
- Custom function apply ki

🗓️ Week 4 — Business Analysis Project (IMPORTANT 🔥)

Ab ek real-world style project karna hai.

Project Ideas

- E-commerce sales analysis
- Customer behavior analysis

- Food delivery data analysis
- Employee attrition analysis

Project Structure

1️⃣ Data cleaning
2️⃣ SQL analysis
3️⃣ Pandas deep analysis
4️⃣ Visualizations
5️⃣ Business insights (MOST IMPORTANT)

Example insight style:

- “Repeat customers generate 60% revenue”
- “Weekend sales highest”

✅ Checklist (Week 4)

- SQL queries project me use ki
- Pandas + visualization done
- Insight section likha
- GitHub repo update kiya

📦 Month-2 Deliverable (GitHub)

SQL-Pandas-Business-Analysis

│

├── data/

├── sql_queries.sql

├── analysis_notebook.ipynb

```
├── visuals/
└── README.md
```

🧠 Daily Routine (best combo)

अगर 2–3 hrs daily:

- 30 min SQL practice
- 60 min Pandas work
- 30 min project progress
- 15 min revision

⚠️ Common Mistakes (avoid)

❌ Sirf Python ya sirf SQL
❌ Copy-paste notebook
❌ Insight explain na karna

✔ Business language me likho.

🔥 Month-2 End Reality Check

Agar ye complete ho gaya to:

➡️ Tum **70% freshers se ahead** ho already.

🎯 Month-3 Goal (clear target)

Month end tak tum:

✔ Machine Learning ka basic flow samajh jao
✔ Data ko train/test me split kar pao
✔ 3–4 ML algorithms use kar pao
✔ Model evaluate kar pao
✔ First ML project GitHub pe upload karo

📅 Week-by-Week Plan

🗓️ Week 1 — ML Basics (Mindset Build)

Sabse pehle ML ka logic samajhna zaroori hai.

Topics

- What is ML (Supervised vs Unsupervised)
- Features vs Target
- Training vs Testing data
- Train/Test split

Tools:

- Scikit-learn (core library)

✅ Checklist (Week 1)

- ML workflow samajh liya
- Dataset split kar pa rahe ho
- Feature & target identify kar pa rahe ho

- First simple model run kiya

🗓️ Week 2 — Regression Models (Prediction start)

Regression = numbers predict karna.

Topics

- Linear Regression
- Multiple Regression
- Feature impact understanding

Practice examples:

- House price prediction
- Salary prediction

Focus:
✔ Interpretation (sirf accuracy nahi)

✅ Checklist (Week 2)

- Linear regression run ki
- Prediction samjha
- Graph me result visualize kiya
- Model results explain kiya

🗓️ Week 3 — Classification Models (REAL WORLD USE)

Classification = categories predict karna.

Topics

- Logistic Regression
- Decision Tree
- Random Forest

Practice examples:

- Customer churn
- Loan approval
- Spam detection

Important Metrics

- Accuracy
- Precision
- Recall
- F1 Score

✅ Checklist (Week 3)

- 2 classification models banaye
- Metrics compare ki
- Overfitting concept samjha
- Feature importance dekha

🗓️ Week 4 — First End-to-End ML Project 🔥

Ab sab combine karna hai.

Project Ideas (choose 1)

- Customer Churn Prediction
- Credit Risk Prediction
- Employee Attrition Analysis
- Sales Forecasting

Project Flow (VERY IMPORTANT)

1️⃣ Data cleaning
2️⃣ EDA (Month-1 skills)
3️⃣ Feature selection
4️⃣ ML model training
5️⃣ Evaluation
6️⃣ Insights + conclusion

✅ Checklist (Week 4)

- End-to-end ML pipeline banayi
- Multiple models compare kiye
- Best model choose kiya
- GitHub repo upload

📦 Month-3 Deliverable (GitHub Structure)

```
ML-Project-Month3
│
├── data/
├── notebook.ipynb
```

```
├── model_results/
├── requirements.txt
└── README.md
```

🧠 Daily Routine (best format)

अगर 2–3 hrs daily:

- 40 min theory (concept samajhna)
- 60 min coding
- 30 min experiment (parameters change karo)

⚠️ Biggest Mistakes (avoid)

❌ Accuracy ke peeche bhaagna
❌ Sirf code run karna
❌ Model explain na kar pana

✔ Interview me puchte hain:
"Why this model?"

🔥 Month-3 End Reality Check

Agar ye complete ho gaya:

➡️ Tum officially **beginner data analyst nahi**,
entry-level ML engineer mindset me aa gaye ho.

🎯 Month-4 Goal (clear target)

Month end tak tum:

✔ Model improve karna jaan jao
✔ Overfitting fix kar pao
✔ Cross-validation use kar pao
✔ Hyperparameter tuning kar pao
✔ Industry-level models (Random Forest / XGBoost) use kar pao

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📅 Week-by-Week Plan

🗓️ Week 1 — Model Evaluation Deep Dive

Ab sirf accuracy nahi — model quality samajhni hai.

Topics

- Bias vs Variance
- Overfitting vs Underfitting
- Confusion Matrix
- ROC-AUC curve (basic understanding)

Practice

Same dataset pe:

- Simple model vs complex model compare karo.

✅ Checklist (Week 1)

- Overfitting identify kar pa rahe ho

- Confusion matrix samajh li
- Multiple metrics compare kiye
- Train vs test performance check ki

🗓️ Week 2 — Cross Validation (VERY IMPORTANT)

Interview favourite topic 😄

Topics

- K-Fold Cross Validation
- Why CV better than single split

Practice:

- sklearn cross_val_score()

Goal

Model stable hai ya nahi — ye samajhna.

✅ Checklist (Week 2)

- K-fold run kiya
- CV score samjha
- Stable model identify kiya
- Different splits test kiye

🗓️ Week 3 — Hyperparameter Tuning (REAL DS SKILL)

Model ko smart banana.

Topics

- GridSearchCV
- RandomizedSearchCV
- Important parameters (depth, estimators etc.)

Models Focus

- Random Forest
- Decision Tree

✅ Checklist (Week 3)

- Grid search run kiya
- Best params nikale
- Performance improve ki
- Before vs after compare kiya

🗓️ Week 4 — Advanced Models (Industry Favourite 🔥)

Ab serious model use karte hain.

Learn

- Random Forest (deep understanding)
- XGBoost / Gradient Boosting (basic practical)

Reason:
Companies me frequently use hota hai.

End-to-End Project (Month-4 Main Task)

Project idea:

- Customer churn advanced prediction
- Fraud detection
- Sales forecasting advanced

Include:
✔ Baseline model
✔ Tuned model
✔ Performance comparison
✔ Final insight

✅ Checklist (Week 4)

- XGBoost run kiya
- Model tuning apply ki
- Results compare kiye
- Final report likha

📦 Month-4 GitHub Deliverable

```
Advanced-ML-Optimization
│
├── data/
├── baseline_model.ipynb
├── tuned_model.ipynb
├── results_comparison.csv
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs:

- 30 min concept
- 60 min coding
- 30 min experiment (parameters change)

⚠️ Common Mistakes (avoid)

❌ Random parameters change karna
❌ Over-optimization (data leak)
❌ Results explain na karna

✔ Always ask:
"Business value kya improve hua?"

🔥 Month-4 End Reality Check

Agar ye complete ho gaya to:

➡️ Tum basic DS nahi — **strong ML practitioner** level pe ho.

Ye stage pe many internships crack ho sakte hain.

🎯 Month-5 Goal (clear target)

Month end tak tum:

✔ Real-world messy data handle kar pao
✔ Strong feature engineering kar pao
✔ ML pipelines banana samajh jao
✔ Data leakage avoid karna seekh lo
✔ Professional workflow use karo

📅 Week-by-Week Plan

🗓️ Week 1 — Feature Engineering Basics (GAME CHANGER)

Ab model ko smart banane ka kaam start.

Topics

- Feature creation (new columns banana)
- Date-time features
- Interaction features
- Binning (age groups etc.)

Example:

purchase_date → month / weekday / weekend

Practice

Same dataset pe naye features banao aur performance compare karo.

✅ Checklist (Week 1)

- 5+ naye features create kiye
- Date features use kiye
- Before vs after result compare kiya
- Feature usefulness samjha

🗓️ Week 2 — Categorical & Missing Data Handling

Real datasets clean nahi hote 😄

Topics

- Label Encoding
- One-Hot Encoding
- Rare categories handling
- Missing value strategies:
    - Mean / Median
    - Mode
    - Advanced fill logic

Practice

Dataset intentionally mess karo → phir clean karo.

✅ Checklist (Week 2)

- Encoding implement ki
- Missing values strategy test ki
- Model performance check ki
- Wrong encoding ka effect samjha

🗓️ Week 3 — Pipelines (PRO LEVEL SKILL)

Industry me manual steps nahi chalte.

Topics

- sklearn Pipeline
- ColumnTransformer
- Full preprocessing + model flow

Example flow:

Clean → Encode → Scale → Model

Why important?

Interview me pipeline use = strong candidate signal.

✅ Checklist (Week 3)

- First pipeline banayi
- Multiple preprocessing steps add kiye
- Pipeline ke through model train kiya
- Code clean ho gaya

🗓️ Week 4 — Realistic Project (IMPORTANT 🔥)

Ab ek **messy real dataset** choose karo.

Project Ideas

- Customer lifetime value prediction
- Marketing campaign success
- Employee performance analysis
- E-commerce purchase behavior

Project Flow

1️⃣ Raw messy data
2️⃣ Heavy feature engineering
3️⃣ Pipeline setup
4️⃣ Tuned model
5️⃣ Insights

✅ Checklist (Week 4)

- Feature engineering applied
- Pipeline use ki
- Model train + evaluate kiya
- GitHub project complete

📦 Month-5 Deliverable (GitHub)

```
Feature-Engineering-Pipeline-Project
│
├── data/
├── preprocessing_pipeline.ipynb
├── model_training.ipynb
├── pipeline.pkl
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs/day:

- 30 min concept
- 60 min experiment
- 30 min notebook cleanup

⚠️ Common Mistakes (VERY COMMON)

❌ Random features banana
❌ Train data se test data leak kar dena
❌ Pipeline use na karna

✔ Always remember:
Feature engineering = business understanding.

🔥 Month-5 End Reality Check

Agar ye month complete ho gaya:

➡️ Tum **80% entry-level candidates se ahead** ho.

Because most लोग:

- model chalana jaante hain
- but data prepare nahi.

🎯 Month-6 Goal (clear target)

Month end tak tum:

✔ ML model ko API bana sako
✔ Model save / load kar pao
✔ Basic deployment samajh jao
✔ Production mindset develop ho
✔ End-to-end usable ML app build karo

📅 Week-by-Week Plan

🗓️ Week 1 — Model Saving & Production Basics

First step: model ko notebook ke bahar le jana.

Topics

- joblib / pickle
- Model serialization
- Inference vs Training difference
- Input validation basics

Practice

Existing model save karo → phir alag file me load karo.

✅ Checklist (Week 1)

- Model save kiya (.pkl file)
- Model load successfully

- New data pe prediction ki
- Training vs inference samjha

🗓️ Week 2 — FastAPI / Flask (API Build 🔥)

Industry standard skill.

Topics

- FastAPI basics (recommend)
- API endpoints
- JSON input/output
- POST request

Goal

Model ko API bana do:

Input → API → Prediction

✅ Checklist (Week 2)

- FastAPI install & setup
- Prediction endpoint banaya
- JSON input accept kar raha
- API local run ho rahi

🗓️ Week 3 — Production Thinking (VERY IMPORTANT)

Ye stage most learners skip kar dete hain.

Topics

- Error handling
- Input validation
- Logging basics
- Versioning mindset

Practice

Wrong input bhejo → API crash na kare.

✅ Checklist (Week 3)

- Input validation add ki
- Error handling use ki
- Clean API response
- Code structure improve ki

🗓️ Week 4 — End-to-End Deployable Project 🚀

Ab real project banao.

Project Ideas

- Loan approval prediction API
- Churn prediction API
- Price prediction service
- Customer risk scoring API

Full Flow (MUST FOLLOW)

1️⃣ Data preprocessing
2️⃣ Model training
3️⃣ Model save
4️⃣ API creation
5️⃣ Prediction response

✅ Checklist (Week 4)

- Model + API integrated
- End-to-end run ho raha
- README with usage
- GitHub pe complete project

📦 Month-6 GitHub Deliverable

```
ML-API-Deployment
│
├── app.py
├── model.pkl
├── requirements.txt
├── train_model.ipynb
└── README.md
```

🧠 Daily Routine (ideal)

अगर 2–3 hrs/day:

- 30 min concept
- 60 min coding
- 30 min testing API

⚠️ Common Mistakes (avoid)

❌ Notebook code directly API me daal dena
❌ Input validation skip
❌ README na likhna

✔ Remember:
Recruiters LOVE deployable projects.

🔥 Month-6 End Reality Check

Agar ye complete ho gaya:

➡️ Tum sirf Data Scientist nahi,
AI/Product Engineer mindset me enter kar chuke ho.

Ye hi profile AI companies ko pasand aati hai.

🔥 **Deep Learning + Neural Networks**

Ye month foundation banayega future LLM / GenAI skills ke liye.

🎯 Month-7 Goal (clear target)

Month end tak tum:

✔ Neural Networks ka intuition samajh jao
✔ Deep learning basics implement kar pao
✔ PyTorch use karna start kar do
✔ Classification deep model bana pao
✔ First deep learning project complete karo

📅 Week-by-Week Plan

🗓️ Week 1 — Neural Network Basics (Mindset Shift)

Sabse pehle samajhna hai *kaise kaam karta hai*.

Topics

- Artificial Neural Network (ANN) idea
- Layers (input, hidden, output)
- Activation functions (ReLU, Sigmoid)
- Forward pass vs backpropagation (intuition level)

⚠️ Math deep nahi — concept samjho.

Practice

Simple dataset pe ANN run karo.

✅ Checklist (Week 1)

- Neural network flow samjha
- Activation functions idea clear
- First ANN model run kiya
- Loss ka concept samjha

🗓️ Week 2 — PyTorch Basics (VERY IMPORTANT)

Deep learning ka real skill = framework use.

Topics

- Tensors
- Dataset & DataLoader
- Model class banana
- Training loop basics

Practice

Small classification dataset.

✅ Checklist (Week 2)

- Tensor operations kiye
- PyTorch model class likhi
- Training loop run kiya
- Loss decrease hota dekha 😄

🗓️ Week 3 — Model Training Deep Dive

Ab model improve karna seekho.

Topics

- Epochs & batch size
- Learning rate
- Optimizers (Adam, SGD)
- Validation data

Experiment

Different learning rate try karo.

✅ Checklist (Week 3)

- Optimizer change karke test kiya
- Overfitting observe ki
- Train vs validation compare ki
- Model tuning basics samjhe

🗓️ Week 4 — Deep Learning Project 🔥

Ab ek real DL project karo.

Project Ideas

- Image classification (MNIST / Fashion MNIST)
- Customer classification deep model

- Tabular data neural network

Project Flow

1️⃣ Data preparation
2️⃣ Model build
3️⃣ Train + validation
4️⃣ Performance metrics
5️⃣ Results explanation

✅ Checklist (Week 4)

- End-to-end DL project complete
- PyTorch training loop use ki
- Accuracy evaluate ki
- GitHub upload

📦 Month-7 Deliverable (GitHub)

```
Deep-Learning-Basics
│
├── data/
├── model.py
├── train.py
├── notebook.ipynb
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs/day:

- 30 min concepts

- 60 min coding
- 30 min experiments

DL me experimentation = learning.

⚠️ Common Mistakes (bahut common)

❌ Deep math me fas jana
❌ Sirf copy-paste tutorials
❌ Model run kiya but samjha nahi

✔ Always ask:
"Ye parameter change karne se kya hua?"

🔥 Month-7 End Reality Check

Agar ye complete ho gaya:

➡️ Tum **ML se AI zone** me enter kar chuke ho.

Ab next months me:

- LLMs
- GenAI
- RAG systems

samajhna easy ho jayega.

Main honestly bolu — yahi month recruiters ko impress karta hai, kyunki ab tum real AI models use karoge.

🎯 Month-8 Goal (clear target)

Month end tak tum:

✔ CNN samajh jao (image AI backbone)
✔ Transfer Learning use kar pao
✔ Pretrained models ka use samjho
✔ Real image-based AI project complete karo
✔ Deep learning workflow strong ho

📅 Week-by-Week Plan

🗓️ Week 1 — CNN Basics (Image AI Start)

CNN = Computer Vision ka foundation.

Topics

- Convolution layer
- Filters / kernels (intuition level)
- Pooling layers
- Feature extraction idea

💡 Samjho:
CNN image ke patterns detect karta hai.

Practice

MNIST / CIFAR-10 dataset use karo.

✅ Checklist (Week 1)

- CNN concept samjha
- Simple CNN model run kiya
- Image classification result dekha
- CNN vs ANN difference samjha

📅 Week 2 — CNN Training & Improvement

Ab model improve karna.

Topics

- Data augmentation
- Dropout
- Batch normalization (basic idea)
- Overfitting in deep learning

Practice

Model accuracy improve karne ki कोशिश करो.

✅ Checklist (Week 2)

- Data augmentation apply ki
- Overfitting reduce ki
- Validation accuracy improve ki
- Model training samjha

📅 Week 3 — Transfer Learning (🔥 INDUSTRY MUST)

REAL AI engineer skill.

Kyun?
➡️ Companies mostly pretrained models use karti hain.

Topics

- Pretrained models concept
- Feature extractor vs fine-tuning
- Models:
    - ResNet
    - MobileNet
    - EfficientNet (basic intro)

Practice

Pretrained model use karke image classification karo.

✅ Checklist (Week 3)

- Pretrained model load kiya
- Last layers modify ki
- Fine-tuning ki
- Training faster samjha

🗓️ Week 4 — Real AI Project (Portfolio Level 🔥)

Ab ek serious project banao.

Project Ideas (choose 1)

- Waste classification AI
- Food image classifier
- Animal breed classifier
- Disease detection from leaf images
- Face mask detection

Project Structure

1️⃣ Data preparation
2️⃣ Transfer learning
3️⃣ Training + evaluation
4️⃣ Prediction demo
5️⃣ Results explanation

✅ Checklist (Week 4)

- Transfer learning use ki
- End-to-end project complete
- Accuracy properly evaluate
- GitHub upload with README

📦 Month-8 Deliverable (GitHub)

```
Computer-Vision-AI
│
├── data/
```

```
├── model.py
├── train.py
├── predict.py
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs/day:

- 30 min concept
- 60 min training experiments
- 30 min model tuning

⚠️ Common Mistakes (avoid)

❌ Scratch se model banana (time waste)
❌ Transfer learning ignore karna
❌ GPU use na karna (slow learning)

✔ Industry rule:
Pretrained model first.

🔥 Month-8 End Reality Check

Agar complete ho gaya:

➡️ Tum officially **AI Engineer level beginner** ho.

Seriously — is stage pe portfolio already strong lagta hai.

🎯 Month-9 Goal (clear target)

Month end tak tum:

✔ NLP basics samajh jao
✔ Text preprocessing kar pao
✔ Transformers ka basic idea clear ho
✔ Pretrained NLP models use kar pao
✔ First text-based AI project bana lo

📅 Week-by-Week Plan

🗓️ Week 1 — NLP Basics (Foundation)

Text data normal data se alag hota hai.

Topics

- Tokenization
- Stopwords
- Stemming / Lemmatization
- Bag of Words
- TF-IDF (intuition level)

Practice

Simple sentiment analysis dataset.

✅ Checklist (Week 1)

- Text clean karna aaya

- Tokenization samjha
- TF-IDF use kiya
- Basic text model run kiya

🗓️ Week 2 — Classical NLP Models

Deep learning se pehle basics.

Topics

- Naive Bayes
- Logistic Regression for text
- Text classification pipeline

Practice ideas

- Movie review sentiment
- Spam detection

✅ Checklist (Week 2)

- Text classification project banaya
- Accuracy evaluate ki
- Feature extraction samjha
- Baseline model ready

🗓️ Week 3 — Transformers Introduction (🔥 GAME CHANGE)

Ab modern NLP.

Topics

- What is Transformer (high-level)
- Attention concept (basic intuition)
- Pretrained models idea
- Hugging Face ecosystem intro

Practice

Pretrained model load karo aur inference run karo.

Example tasks:

- Sentiment analysis
- Text summarization
- Question answering

✅ Checklist (Week 3)

- Transformer concept samjha
- Hugging Face use kiya
- Pretrained model run kiya
- Output samjha

🗓️ Week 4 — NLP Project (Portfolio Level 🔥)

Ab real project banao.

Project Ideas (choose 1)

- Product review sentiment analyzer
- News classification AI
- Resume classifier
- Chatbot intent detection
- Text summarizer

Project Flow

1️⃣ Data preprocessing
2️⃣ Baseline model
3️⃣ Transformer model
4️⃣ Performance compare
5️⃣ Insights

✅ Checklist (Week 4)

- Transformer use kiya
- End-to-end NLP pipeline
- Result comparison kiya
- GitHub upload

📦 Month-9 Deliverable (GitHub)

```
NLP-Transformer-Project
│
├── data/
├── preprocessing.ipynb
├── transformer_model.ipynb
```

```
├── inference.py
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs/day:

- 30 min concept
- 60 min coding
- 30 min experiment (different models)

⚠️ Common Mistakes (avoid)

❌ Transformer ka deep math me fas jana
❌ Model copy-paste karna bina samjhe
❌ Text cleaning skip karna

✔ Focus:
Pipeline samjho.

🔥 Month-9 End Reality Check

Agar ye complete hua:

➡️ Tum **LLM-ready stage** me aa gaye ho.

Honestly — ab tum modern AI tools samajh sakte ho.

🔥 Portfolio + Open Source + LLM Project = Industry-ready identity

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🎯 Month-10 Goal (clear target)

Month end tak tum:

✔ LLM + GenAI practical use samajh jao
✔ RAG system bana pao
✔ Real AI assistant project complete karo
✔ Open source contribution visible ho
✔ Job-ready portfolio finish karo

📅 Week-by-Week Plan

🗓️ Week 1 — LLM Practical Basics (No Theory Overload)

Ab ChatGPT-type systems samjho.

Topics

- LLM workflow (high level)
- Embeddings concept
- Vector database idea
- Prompt engineering basics

Goal:
➡️ Model use karna, train nahi.

Practice

- Text summarize

- Q&A system
- Prompt experiments

✅ Checklist (Week 1)

- Embeddings concept clear
- Prompt experimentation ki
- API-based LLM use kiya
- Output quality compare ki

🗓️ Week 2 — RAG (Retrieval Augmented Generation) 🔥

Yeh modern AI ka real skill hai.

Topics

- Documents → embeddings
- Vector store basics
- Retrieve + Generate flow

Simple idea:

Docs → Search → LLM → Answer

Practice Project

PDF / notes se Q&A bot.

✅ Checklist (Week 2)

- Documents chunk kiye
- Embeddings create ki
- Retrieval system banaya
- Context-aware answer mila

🗓️ Week 3 — Open Source Contribution (VERY IMPORTANT)

Ab tum beginner nahi ho — contribute start karo.

Where to contribute

- Hugging Face repos
- LangChain docs/issues
- Scikit-learn docs fixes
- FastAPI examples

Start Strategy (REALISTIC)

1️⃣ “good first issue” search
2️⃣ Documentation improvement
3️⃣ Small bug fix or example addition

✅ Checklist (Week 3)

- 1 open-source repo choose kiya
- First PR raise ki
- Maintainer feedback mila
- GitHub activity visible

🗓️ Week 4 — FINAL CAPSTONE PROJECT 🚀

Ab 10 month ka FINAL project.

🔥 Project Idea (Strongest Option)

AI Knowledge Assistant (RAG based)

Example:

- Company docs assistant
- Study notes chatbot
- Policy Q&A AI
- Research assistant

Must Include

✔ Data ingestion
✔ Embeddings
✔ Retrieval pipeline
✔ LLM response
✔ Simple UI / API

Bonus (extra strong profile)

- Deploy on cloud (basic)
- Logging & response improvement

✅ Checklist (Week 4)

- End-to-end RAG system
- Clean project structure

- README with architecture
- Demo screenshot/video
- GitHub polished

📦 Final Month GitHub Deliverable

```
LLM-RAG-AI-Assistant
│
├── data/
├── embeddings/
├── app.py
├── rag_pipeline.py
├── requirements.txt
└── README.md
```

🧠 Daily Routine (best)

अगर 2–3 hrs/day:

- 30 min learning
- 60 min building
- 30 min debugging / improvement

⚠️ Common Mistakes (avoid)

❌ LLM ko magic samajhna
❌ Sirf prompts likhna
❌ Architecture explain na kar paana

✔ Interview rule:
Explain **HOW** system works.

🔥 FINAL REALITY CHECK (after Month-10)

Agar tumne honestly complete kiya:

➡️ Tumhara profile ho jayega:

⭐ Data Scientist
⭐ ML Engineer
⭐ Beginner AI / LLM Engineer

Aur sabse important:

> 💥 Tum “course learner” nahi — **AI builder** lagoge.

🚀 Ab ek SECRET bataun? (honest)

10 month ke baad top 5% fast growth log ye karte hain:

✔ Regular GitHub commits
✔ Open source continue
✔ AI project improve karte rehte